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Information processing device, information processing method, and storage medium

Abstract

An information processing device that is usable as a key of a vehicle, the information processing device including: a display section that is configured to display information of at least one other device that is connected to the information processing device; a memory; and a processor coupled to the memory and that is configured to: receive a user selection selecting, from the at least one other device displayed at the display section, at least one other device to be operated as a key of the vehicle; and execute duplication processing to each of the selected at least one other device to allow the selected at least one other device to be used as the key of the vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-100568 filed on Jun. 22, 2022, the disclosure of which is incorporated by reference herein.

BACKGROUND

Technical Field

(2) The present disclosure relates to an information processing device, an information processing method, and a non-transitory storage medium storing an information processing program.

Related Art

(3) Cars are widely used as an element of mobility services, and various services such as car sharing are expected to become more numerous and commonplace. In accordance with this trend, smartphones are being used as keys for cars, and demands on wireless digital key systems that do not require the handover of a physical key are rising.

(4) Technologies relating to wireless digital key systems include, for example, Japanese Patent Application Laid-Open (JP-A) No. 2020-033707. JP-A No. 2020-033707 discloses an electronic key system for a vehicle that is provided with a portable device and an on-board apparatus. The electronic key system unlocks doorlocks of the vehicle and suchlike, dependent on authentication of the portable device by the on-board apparatus.

(5) The Car Connectivity Consortium (CCC) defines a standard for electronic key systems for vehicles. In the standard defined by CCC, devices that are an owner device and friend devices are defined. The owner device is a device used by an owner of a vehicle, and the friend devices are devices that are expected to be used by family members or friends of the owner.

(6) However, when plural friend devices are registered in addition to an owner device conforming to CCC, additionally registering the friend devices one at a time is effortful for a user.

SUMMARY

(7) The present disclosure provides an information processing device, an information processing method and a non-transitory storage medium storing an information processing program that, when other terminals are added as a key of a vehicle in addition to a terminal that is used as the key, reduce the effort of registering the additional terminals.

(8) A first aspect of the present disclosure is an information processing device that is usable as a key of a vehicle, the information processing device including: a display section that is configured to display information of at least one other device that is connected to the information processing device; memory; and a processor coupled to the memory and that is configured to: receive a user selection selecting, from the at least one other device displayed at the display section, at least one other device to be operated as a key of the vehicle; and execute duplication processing to each of the selected at least one other device to allow the selected at least one other device to be used as the key of the vehicle.

(9) The information processing device according to the first aspect duplicates key information from that device to the other device(s), enabling use of the other device(s) as the key of the vehicle. In this information processing device, at least one other device to be operated as the key of the vehicle is selected by a user from the at least one other device displayed at the display section. The duplication processing allowing the other device(s) to be used as the key of the vehicle is executed to each of the selected at least one other device. According to this information processing device, when other terminals are added as a key of the vehicle in addition to a terminal that is used as the

key, the effort of registering the addition may be reduced.

(10) A second aspect of the present disclosure is an information processing method including: displaying information of at least one other device connected to an information processing device, the information processing device being usable as a key of a vehicle; receiving a user selection selecting, from the displayed at least one other device, at least one other device to be operated as a key of the vehicle; and executing duplication processing to each of the selected at least one other device to allow the selected at least one other device to be used as the key of the vehicle.

(11) A third aspect of the present disclosure is a non-transitory storage medium storing a program that is executable by a computer to execute information processing, the information processing including: displaying information of at least one other device connected to an information processing device, the information processing device being usable as a key of a vehicle; receiving a user selection selecting, from the displayed at least one other device, at least one other device to be operated as a key of the vehicle; and executing duplication processing to each of the selected at least one other device to allow the selected at least one other device to be used as the key of the vehicle.

(12) According to the present disclosure, when other terminals are added as a key of a vehicle in addition to a terminal that is used as the key, effort of registering the additional terminals may be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram showing schematic structures of a key management system according to a first exemplary embodiment.

(2) FIG. 2 is a block diagram showing hardware structures of a vehicle according to the first exemplary embodiment.

(3) FIG. 3 is a block diagram showing hardware structures of a terminal according to the first exemplary embodiment.

(4) FIG. 4 is a block diagram showing functional structures of the terminal according to the first exemplary embodiment.

(5) FIG. 5 is a flowchart showing a flow of registration processing at a terminal serving as an owner device according to the first exemplary embodiment.

(6) FIG. 6 is a diagram showing an example of an image of terminal names and communication systems that is displayed at a touch panel of the terminal serving as the owner device according to the first exemplary embodiment.

(7) FIG. 7 is a diagram showing an example of an image prompting switching of a communication system that is displayed at the touch panel of a terminal serving as a friend device according to the first exemplary embodiment.

(8) FIG. 8 is a flowchart showing a flow of duplication processing at the terminal serving as the owner device according to the first exemplary embodiment.

(9) FIG. 9 is a flowchart showing a flow of response processing at the terminal serving as the friend device according to the first exemplary embodiment.

(10) FIG. 10 is a diagram showing an example of an image requesting approval that is displayed at a touch panel of the terminal serving as the friend device according to the first exemplary embodiment.

(11) FIG. 11 is a diagram showing an example of an image of terminal names that is displayed at a touch panel of a terminal serving as an owner device according to a second exemplary embodiment.

DETAILED DESCRIPTION

(12) A key management system that includes a terminal serving as an information processing device of the present disclosure is described. The key management system is a system that manages electronic keys (below referred to simply as keys) to be used for unlocking, locking, starting and the like of a vehicle.

First Exemplary Embodiment

(13) —Overall Structure—

(14) As shown in FIG. 1, a key management system **10** according to a first exemplary embodiment includes a vehicle **12**, terminals **14** and servers **16**. The servers **16** include a vehicle server **16A** that manages the vehicle **12** and terminal servers **16B** that manage the terminals **14**. The terminals **14** according to the present exemplary embodiment include a terminal **14A** that is an owner device, at which key information is specified in advance, and terminals **14B** that are friend devices at which the key information of the terminal **14A** is duplicated. Each terminal **14** can be illustrated by a terminal such as a smartphone or the like that is carried by a driver of the vehicle **12**. The terminal **14A** is an example of the information processing device and each terminal **14B** is an example of an other device.

(15) The vehicle **12** and the vehicle server **16A** are connected to be capable of communications with one another. Each terminal **14** and the corresponding terminal server **16B** are connected to be capable of communications with one another. Each terminal **14** and the vehicle server **16A** may be connected to be capable of communications with one another. The vehicle server **16A** and terminal servers **16B** are connected to one another via a public network N.

(16) Numbers of the vehicle **12**, the terminals **14** and the servers **16** not limited to the numbers illustrated in FIG. 1. The terminal servers **16B** are not necessarily provided one for each of the terminals **14**; the plural terminals **14** may be connected to one of the terminal server **16B**.

(17) —Vehicle—

(18) As shown in FIG. 2, the vehicle **12** includes a verification electronic control unit (ECU) **20**, communication sections **22**, a body ECU **24** and an engine ECU **26**.

(19) The verification ECU **20** includes a central processing unit (CPU) **20A**, read-only memory (ROM) **20B**, random access memory (RAM) **20C**, an in-vehicle communications interface **20D** and a wireless communications interface **20E**. The CPU **20A**, ROM **20B**, RAM **20C**, in-vehicle communications interface **20D** and wireless communications interface **20E** are connected to be capable of communications with one another via an internal bus (not shown in the drawings).

(20) The CPU **20A** is a central arithmetic processing unit that executes various programs and controls respective parts. That is, the CPU **20A** reads a program from the ROM **20B** and executes the program, using the RAM **20C** as a workspace.

(21) The ROM **20B** stores various programs and various kinds of data. Key information **100**, which is information relating to a key of the vehicle **12**, is stored at the ROM **20B** according to the present exemplary embodiment.

(22) The RAM **20C** serves as a work area and temporarily stores programs and data.

(23) The in-vehicle communications interface **20D** is an interface for connecting with a body ECU **24** and an engine ECU **26**. The interface uses a communications standard according to Controller Area Network (CAN) or Ethernet (registered trademark). The in-vehicle communications interface **20D** is connected to an external bus **28**.

(24) The wireless communications interface **20E** is an interface for communicating with the vehicle server **16A**. The wireless communications interface **20E** employs a communications standard such as 5G, 4G, LTE or the like. If communication with the vehicle server **16A** by a data communication module (DCM), another ECU or the like is possible, the wireless communications interface **20E** need not necessarily be included at the verification ECU **20**.

(25) The body ECU **24** is an ECU that controls equipment such as lights, doors and the like of the vehicle **12**. The body ECU **24** is connected to a door locking device, which is not shown in the drawings, and may control locking and unlocking of doors of the vehicle **12**.

(26) The engine ECU **26** is an ECU that controls an engine of the vehicle **12**. The engine ECU **26** may control starting and stopping of the engine.

(27) The communication sections **22** include a near field communication (NFC) communication section **22A**, an NFC communication section **22B**, a Bluetooth (registered trademark) Low Energy (BLE) communication section **22C** and an Ultra Wide Band (UWB) communication section **22D**. The NFC communication section **22A** is provided at each door of the vehicle **12**, and the NFC communication section **22B** is provided at a center console or the like of the vehicle **12**. The BLE communication section **22C** and UWB communication section **22D** are disposed at locations that enable reception of electromagnetic waves from inside the vehicle and from surroundings of the vehicle **12**. Plural numbers of the BLE communication section **22C** and the UWB communication section **22D** may be disposed at the vehicle **12**.

(28) —Terminal—

(29) As shown in FIG. 3, each terminal **14** according to the present exemplary embodiment includes a control section **40**, communication sections **42**, and a touch panel **44**.

(30) The control section **40** includes a CPU **40A**, ROM **40B**, RAM **40C** and a communications interface **40D**. The CPU **40A**, ROM **40B**, RAM **40C** and communications interface **40D** are connected to be capable of communications with one another via an internal bus (not shown in the drawings). Functions of the CPU **40A**, ROM **40B**, RAM **40C** and communications interface **40D** are similar to functions of the CPU **20A**, ROM **20B**, RAM **20C** and wireless communications interface **20E** of the verification ECU **20**. In the control section **40** according to the present exemplary embodiment, by executing an execution program **150**, the CPU **40A** functions as a selection section **200**, an execution section **210** and a notification section **220**, which are shown in FIG. 4.

(31) As shown in FIG. 3, the ROM **40B** according to the present exemplary embodiment stores the execution program **150** and key information **160**, which is information on a key of the vehicle **12**. The execution program **150** is a program for implementing registration processing, duplication processing and response processing, which are described below. The key information **160** is data for verification with the key information **100** that is stored at the vehicle **12**.

(32) The touch panel **44** includes a liquid crystal display that serves as a display section and a touch sensor that serves as an input section.

(33) The communication sections **42** include an NFC communication section **42A**, a BLE communication section **42C**, a UWB communication section **42D** and a Wi-Fi® communication section **42E**. The NFC communication section **42A**, BLE communication section **42C** and UWB communication section **42D** are examples of a first communication section, and the Wi-Fi communication section **42E** is an example of a second communication section.

(34) As shown in FIG. 4, the selection section **200** features a function allowing a user to select which of the terminals **14B** are to be operated as the key of the vehicle **12**. More specifically, the selection section **200** allows a user to select the terminals **14B** that are to operate as the key of the vehicle **12** from among the terminals **14B**, which serve as an other device, displayed at the touch panel **44**.

(35) The execution section **210** features a function that executes duplication processing to each terminal **14B** selected by the selection section **200** in order to allow the terminal **14B** to be used as the key of the vehicle **12**. The duplication processing transmits a duplication request from the terminal **14A**, which is already operating as the key, to the terminal **14B** that is to operate as the key subsequently. When an approval notification is received from the terminal **14B** to which the duplication request has been transmitted, the duplication processing transmits the key information to the terminal **14B** that transmitted the approval notification.

(36) The execution section **210** transmits information to each terminal **14B** identifying whether one or a plural number of the terminals **14B** is implementing the duplication processing. The execution section **210** also transmits information to the terminal **14B** indicating whether or not user approval

at the terminal **14B** is required.

(37) In the present exemplary embodiment, the duplication processing may be executed in a case in which the terminal **14B** that is to be a friend device is connected to the terminal **14A** that is the owner device via the NFC communication section **42A**, the BLE communication section **42C** or the UWB communication section **42D**. On the other hand, in a case in which the terminal **14B** is connected to the terminal **14A** via the Wi-Fi communication section **42E**, the duplication processing cannot be executed. Therefore, in a case in which a terminal **14B** that is connected via the Wi-Fi communication section **42E** is selected by a user, the execution section **210** prompts connection with the NFC communication section **42A**, BLE communication section **42C** or UWB communication section **42D**, and after the terminal **14B** has reconnected via any one of the NFC communication section **42A**, BLE communication section **42C** and UWB communication section **42D**, the execution section **210** executes the duplication processing to that terminal **14B**.

(38) The notification section **220** features a function that, in a case in which the execution section **210** has executed the duplication processing to each selected terminal **14B**, notifies a result of the duplication processing to each terminal **14B** to the user of the terminal **14A** that is the owner device. More specifically, the notification section **220** notifies the results of the duplication processing to the user by a display at the touch panel **44**.

(39) —Flow of Control—

(40) Flows of processing that is executed in the key management system **10** according to the present exemplary embodiment are described using the flowcharts in FIG. 5, FIG. 8 and FIG. 9. The processing at each terminal **14** is implemented by the CPU **40A** functioning as the selection section **200**, the execution section **210** and the notification section **220**. Although a situation in which the CPU **40A** execute the processes shown in the drawings is recited, this is not limiting. The flowcharts described below may be implemented by one or more processors loading and executing programs stored at one or more memories.

(41) First, the registration processing executed at the terminal **14A** that is the owner device is described.

(42) In step **S100** of FIG. 5, the CPU **40A** searches for the terminals **14B** that are connected via the respective communication sections **42**.

(43) In step **S101**, the CPU **40A** executes display processing. More specifically, the CPU **40A** displays each terminal **14B** connected to the terminal **14A** at the touch panel **44**. For example, as shown in FIG. 6, terminal names are displayed at the touch panel **44** together with the communication systems connecting those terminals.

(44) In step **S102** of FIG. 5, the CPU **40A** accepts selection of each terminal **14B** that is to duplicate the key and be a friend device. More specifically, the CPU **40A** accepts selections of the terminals **14B** displayed at the touch panel **44** through user operations on the touch panel **44**.

(45) In step **S103**, the CPU **40A** makes a determination as to whether each terminal **14B** is connected by Wi-Fi. In a case in which the CPU **40A** determines that any of the terminals **14B** is connected by Wi-Fi (Yes in step **S103**), the CPU **40A** proceeds to step **S104**. Alternatively, in a case in which the CPU **40A** determines that no terminal **14B** is connected by Wi-Fi (No in step **S103**) the CPU **40A** proceeds to step **S105**.

(46) In step **S104**, the CPU **40A** transmits a switching request to each terminal **14B** connected by Wi-Fi. Accordingly, the terminal **14B** receiving the switching request displays, for example, a screen at the touch panel **44** thereof that prompts switching of the connection from Wi-Fi to BLE, as illustrated in FIG. 7.

(47) In step **S105** of FIG. 5, the CPU **40A** executes the duplication processing. Details of the duplication processing are described below.

(48) In step **S106**, the CPU **40A** makes a determination as to whether the duplication processing has been attempted to every terminal **14B**. In a case in which the CPU **40A** determines that the processing has been attempted to every terminal **14B** (Yes in step **S106**), the CPU **40A** proceeds to

step **S107**. Alternatively, in a case in which the CPU **40A** determines that the processing has not been attempted to every terminal **14B** (No in step **S106**), the CPU **40A** returns to step **S105**.

(49) In step **S107**, the CPU **40A** notifies processing results for the terminals **14B** that have attempted duplication of the key. That is, for each terminal **14B** that has succeeded in duplicating the key of the terminal **14A**, the CPU **40A** displays a message reporting the success of the duplication at the touch panel **44**, and for each terminal **14B** that has not succeeded in duplicating the key of the terminal **14A**, the CPU **40A** displays a message reporting the failure of the duplication at the touch panel **44**.

(50) In step **S108**, the CPU **40A** makes a determination as to whether the registration processing is to end. For example, the CPU **40A** determines that the registration processing is to end when the user operates an end button displayed at the touch panel **44**. In a case in which the CPU **40A** determines that the registration processing is to end (Yes in step **S108**), the CPU **40A** ends the registration processing. Alternatively, in a case in which the CPU **40A** determines that the registration processing is not to end (No in step **S108**), the CPU **40A** returns to step **S105**. Thus, the registration processing may be executed repeatedly to each terminal **14B** that has failed in duplication of the key.

(51) Now, details of the duplication processing in step **S105** are described.

(52) In step **S200** of FIG. **8**, the CPU **40A** makes a determination as to whether a plural number of the terminals **14B** are to duplicate the key. In a case in which the CPU **40A** determines that plural terminals **14B** are duplicating the key (Yes in step **S200**), the CPU **40A** proceeds to step **S201**. Alternatively, in a case in which the CPU **40A** determines that the terminal **14B** duplicating the key is not plural, that is, is just one terminal (No in step **S200**), the CPU **40A** proceeds to step **S203**.

(53) In step **S201**, the CPU **40A** transmits a multiple duplication request to each terminal **14B** duplicating the key. The meaning of the term “multiple duplication request” as used herein includes a duplication request to which information indicating that the duplication processing is being performed at another of the terminals **14B** is attached. Each terminal **14B** receiving the multiple duplication request may identify that others of the terminals **14B** are also performing the duplication processing. That is, the multiple duplication request is equivalent to “information identifying whether the [terminal **14B** performing the duplication processing] is one device or a plurality of devices”. The terminal **14B** receiving the multiple duplication request may accept approval from a user for performing the duplication processing at the terminals **14B** other than that terminal **14B**. That is, the multiple duplication request is equivalent to “information . . . indicating whether or not user approval from the . . . other device is required”.

(54) In step **S202**, the CPU **40A** makes a determination as to whether an approval notification has been received from every terminal **14B** of the plural terminals **14B**. In a case in which the CPU **40A** determines that the approval notification has been received from every terminal **14B** (Yes in step **S202**), the CPU **40A** proceeds to step **S205**. Alternatively, in a case in which the CPU **40A** determines that the approval notification has not been received from every terminal **14B** (No in step **S202**), the CPU **40A** proceeds to step **S206**.

(55) In step **S203**, the CPU **40A** transmits a duplication request to the terminal **14B** that is duplicating the key.

(56) In step **S204**, the CPU **40A** makes a determination as to whether an approval notification has been received from the terminal **14B**. In a case in which the CPU **40A** determines that the approval notification has been received from the terminal **14B** (Yes in step **S204**), the CPU **40A** proceeds to step **S206**. Alternatively, in a case in which the CPU **40A** determines that the approval notification has not been received from the terminal **14B** (No in step **S204**), the CPU **40A** ends the duplication processing.

(57) In step **S205**, the CPU **40A** transmits key information to every terminal **14B**.

(58) In step **S206**, the CPU **40A** transmits the key information to each terminal **14B** that has approval, that is, each terminal **14B** that has transmitted the approval notification.

(59) In step S207, the CPU 40A receives a duplication result from each terminal 14B to which the CPU 40A transmitted the key information. The CPU 40A then ends the duplication processing.

(60) Now, details of response processing that is executed at each terminal 14B receiving the duplication request is described.

(61) In step S300 of FIG. 9, the CPU 40A receives a duplication request from the terminal 14A.

(62) In step S301, the CPU 40A makes a determination as to whether a plural number of the terminals 14B including the present terminal 14B are duplicating the key. This determination is implemented by determining whether the received duplication request is a multiple duplication request. In a case in which the CPU 40A determines that a plural number of the terminals 14B are duplicating the key (Yes in step S301), the CPU 40A proceeds to step S302. Alternatively, in a case in which the CPU 40A determines that the duplicating terminal 14B is not plural, that is, is just the one terminal 14B (No in step S301), the CPU 40A proceeds to step S305.

(63) In step S302, the CPU 40A accepts an approval default selection. The approval default selection is processing that makes approval selection the default for each other terminal 14B receiving the duplication request. For example, as illustrated in FIG. 10, the approval default selection is accepted by operation of a plural approval button 48C displayed at the touch panel 44 of the terminal 14B.

(64) In step S303 of FIG. 9, the CPU 40A makes a determination as to whether approval default is selected, that is, whether the plural approval button 48C has been operated. In a case in which the CPU 40A determines that approval default is selected (Yes in step S303), the CPU 40A proceeds to step S304. Alternatively, in a case in which the CPU 40A determines that approval default is not selected (No in step S303), the CPU 40A proceeds to step S305.

(65) In step S304, the CPU 40A transmits a representative approval notification to the terminal 14A. The representative approval notification is an approval notification for every terminal 14B.

(66) In step S305, the CPU 40A accepts an approval selection. For example, as illustrated in FIG. 10, the approval selection is accepted by operation of an approval button 48A displayed at the touch panel 44 of the terminal 14B.

(67) In step S306 of FIG. 9, the CPU 40A makes a determination as to whether approval is selected, that is, whether the approval button 48A has been operated. In a case in which the CPU 40A determines that approval is selected (Yes in step S306), the CPU 40A proceeds to step S307. Alternatively, in a case in which the CPU 40A determines that approval is not selected (No in step S306), the CPU 40A proceeds to step S311.

(68) In step S307, the CPU 40A transmits an approval notification to the terminal 14A.

(69) In step S308, the CPU 40A makes a determination as to whether the key information has been received from the terminal 14A. In a case in which the CPU 40A determines that the key information has been received (Yes in step S308), the CPU 40A proceeds to step S309. Alternatively, in a case in which the CPU 40A determines that the key information has not been received (No in step S308), the CPU 40A proceeds to step S310.

(70) In step S309, the CPU 40A executes key specification processing. Thus, the key of the terminal 14A that is the owner device is duplicated at the present terminal 14B.

(71) In step S310, the CPU 40A transmits a duplication result to the terminal 14A. Then the CPU 40A ends the response processing. For the duplication result, a duplication success notification is transmitted to the terminal 14A in a case in which the key specification processing is completed without any problem, and a duplication failure notification is transmitted to the terminal 14A in a case in which there is a problem in the key specification processing and the duplication of the key fails.

(72) In step S311, the CPU 40A transmits an approval denial notification to the terminal 14A. Then the CPU 40A ends the response processing.

(73) —Overview—

(74) The terminal 14A according to the present exemplary embodiment is an information

processing device that is usable as a key of the vehicle **12**, and is configured to execute duplication processing that allows each terminal **14B** selected at the touch panel **44** to be used as the key of the vehicle **12**. That is, in the present exemplary embodiment, the user may set the selected terminals **14B** to be friend devices in a batch at the terminal **14A** that is the owner device. According to the present exemplary embodiment, when plural friend devices are added as the key of the vehicle **12**, the effort of registering the additional devices may be reduced.

(75) Each terminal **14** according to the present exemplary embodiment is configured to include a plural number of the communication sections **42** with different communication systems. The display processing causes the touch panel **44** to display terminal names of the terminals **14B** that may duplicate the key together with the connecting communication systems. In the present exemplary embodiment as described above, each terminal **14B** is formed to be incapable of executing the duplication processing when connected to the terminal **14A** via the Wi-Fi communication section **42E**. Therefore, the registration processing executed at the terminal **14A** prompts each terminal **14B** that is connected only via the Wi-Fi communication section **42E** to reconnect with one of the communication sections **42** that enables execution of the duplication processing. Therefore, according to the present exemplary embodiment, the duplication processing may be performed even if the terminal **14B** has been connected via a communication system with which the duplication processing is not possible.

(76) The duplication processing may first be executed to each terminal **14B** connecting via a communication section **42** other than the Wi-Fi communication section **42E**, and after this duplication processing has ended, each terminal **14B** connecting via the Wi-Fi communication section **42E** may be prompted to reconnect via one of the communication sections **42** that enables execution of the duplication processing.

(77) The terminal **14A** according to the present exemplary embodiment includes the function that, after the duplication processing has been executed at every selected terminal **14B**, reports a result of the duplication processing at each terminal **14B** to the user of the terminal **14A** (see step **S107**). Therefore, according to the present exemplary embodiment, the user may be allowed to select whether to exclude each terminal **14B** at which the duplication processing has failed from the terminals **14B** duplicating the key, or to retry the duplication processing.

(78) The present exemplary embodiment includes the function that transmits the multiple duplication request to each terminal **14B** as information identifying whether one or a plural number of the terminals **14B** is performing the duplication processing (see step **S201**). Therefore, in a case in which it is verified at one of the terminals **14B** that the key is being duplicated at a plural number of the terminals **14B** including that terminal **14B**, approval from the user of each other terminal **14B** may be pre-empted. A setting may be made in advance at the terminal **14A** such that user approval from every terminal **14B** is pre-empted. In this situation, the key information is transmitted to each terminal **14B** without an approval notification being received from the terminal **14B**.

(79) The present exemplary embodiment includes the function that transmits the multiple duplication request to each terminal **14B** as information requesting approval for the performance of the duplication processing at each other terminal **14B** (see step **S201**). Therefore, approval that encompasses each other terminal **14B** may be accepted at one of the terminals **14B**, and the one terminal **14B** may be entrusted with the choice as to whether approval from other users is pre-empted.

(80) As described above, when plural vehicles such as taxis, rental cars or the like are being managed in batches, the present exemplary embodiment is excellent for when adding plural friend devices from the terminal **14A** that is a single owner device.

Second Exemplary Embodiment

(81) In the second exemplary embodiment, the aforementioned display processing of step **S101** is different from the processing according to the first exemplary embodiment. Below, differences

from the first exemplary embodiment are described.

(82) According to the display processing of step **S101**, terminal names of the terminals **14B** connected via the communication sections **42** other than the Wi-Fi communication section **42E** are displayed at the touch panel **44** of the terminal **14A**. For example, as illustrated in FIG. **11**, only the terminal names of terminals that are connected via the BLE communication section **42C** are displayed at the touch panel **44**. In this case, terminal names of the terminals **14B** that are connected via the communication sections **42** other than the BLE communication section **42C** are shown grayed out. Control may be performed such that the terminal names of the terminals **14B** that are connected via the communication sections **42** other than the BLE communication section **42C** are not displayed.

(83) Thus, according to the present exemplary embodiment, terminals for which the duplication processing is possible without changing the communication section **42** that is currently connecting may be displayed, and effort for a user of changing a communication system for the duplication processing may be reduced.

REMARKS

(84) The duplication processing in the exemplary embodiment described above is implemented by direct communications between the terminal **14A** and the terminals **14B**, but this is not limiting. For example, the duplication processing may be implemented via the vehicle server **16A** and terminal servers **16B** or the like. In this case, an effect may be provided in that, when plural friend devices are added as the key of the vehicle **12**, the effort of registering the additional devices may be reduced.

(85) In the exemplary embodiments described above, the CPU **40A** reads and executes software (programs) but the various processes may be executed by various kinds of processor other than a CPU. Examples of processors in these cases include a PLD (programmable logic device) in which a circuit configuration may be modified after manufacturing, such as an FPGA (field programmable gate array) or the like, a dedicated electronic circuit which is a processor with a circuit configuration that is specially designed to execute specific processing, such as an ASIC (application-specific integrated circuit) or the like, and so forth. The processing described above may be executed by one of these various kinds of processors, and may be executed by a combination of two or more processors of the same or different kinds (for example, plural FPGAs, a combination of a CPU with an FPGA, or the like). Hardware structures of these various kinds of processors are, to be more specific, electronic circuits combining circuit components such as semiconductor components and the like.

(86) In the exemplary embodiments described above, modes are described in which programs are stored in advance (installed) on a computer readable non-transitory recording medium. For example, the execution program **150** of the control section **40** is stored in advance at the ROM **40B**. However, this is not limiting and a program may be provided in a mode of being recorded on a non-transitory recording medium such as a CD-ROM (compact disc read-only memory), DVD-ROM (digital versatile disc read-only memory), USB (universal serial bus) memory or the like. Modes are also possible in which a program is downloaded from external equipment via a network.

(87) The flows of processing described in the above exemplary embodiments are examples. Unnecessary steps may be removed, new steps may be added, and processing sequences may be rearranged within a scope not departing from the gist of the disclosure.

Claims

1. An information processing device that is usable as a key of a vehicle, the information processing device comprising: a display section that is configured to display information of at least one other device that is connected to the information processing device; a plurality of communication sections; a memory; and a processor coupled to the memory and that is configured to: receive a

user selection selecting, from the at least one other device displayed at the display section, at least one other device to be operated as a key of the vehicle; and execute duplication processing to each of the selected at least one other device to allow the selected at least one other device to be used as the key of the vehicle; wherein: the plurality of communication sections include a first communication section and a second communication section of a different communication system from the first communication section; the display section is configured to display the at least one other device connected with the any one or more of the communication sections together with a type of a communication system of the communication section connected with the at least one other device; and in a case in which at least one other device that is connected with the second communication section is selected by the user, the processor is configured to prompt connection of the selected at least one other device with the first communication section and, after the selected at least one other device reconnects with the first communication section, execute the duplication processing to the selected at least one other device.

2. The information processing device according to claim 1, wherein the at least one other device displayed at the display section is connected only with any one or more of the communication sections.

3. The information processing device according to claim 1, wherein the processor is configured to, after the duplication processing has been executed to all of the selected at least one other device, report a result of the duplication processing of each of the selected at least one other device to the user.

4. The information processing device according to claim 1, wherein the processor is configured to transmit, to the selected at least one other device for which the duplication processing is performed, information identifying whether the selected at least one other device is one device or a plurality of devices.

5. The information processing device according to claim 1, wherein: the duplication processing includes processing to transmit a duplication request from the information processing device to the selected at least one other device and, after an approval notification from the selected at least one other device to which the duplication request is transmitted is received, transmit information of the key to the selected at least one other device transmitting the approval notification; and the processor is configured to transmit information to the selected at least one other device indicating whether or not user approval from the selected at least one other device is required.

6. The information processing device according to claim 1, wherein the processor is configured to: in a case in which a user selection selecting a plurality of other devices to be operated as keys of the vehicle is received, transmit a multiple duplication request to each of the plurality of other devices, the multiple duplication request indicating that key duplication processing is to be performed for the plurality of other devices; determine whether an approval notification has been received from all of the plurality of other devices; in a case in which the approval notification has been received from all of the plurality of other devices, transmits key information to all of the plurality of other devices; and in a case in which the approval notification has not been received from all of the plurality of other devices, transmit the key information only to one or more devices from which the approval notification has been received.

7. The information processing device according to claim 6, wherein the processor is configured to: determine whether an approval default selection has been received from any one of the plurality of other devices; and in a case in which the approval default selection has been received from any one of the plurality of other devices, the approval notification for all of the plurality of other devices are made default.

8. An information processing method comprising: displaying information of at least one other device connected to an information processing device, the information processing device being usable as a key of a vehicle; receiving a user selection selecting, from the displayed at least one other device, at least one other device to be operated as a key of the vehicle; executing duplication processing to each of the selected at least one other device to allow the selected at least one other

device to be used as the key of the vehicle; displaying the at least one other device connected to the information processing device with any one or more of communication sections together with a type of a communication section connected with the at least one other device; and in a case in which at least one other device that is connected with a second communication section is selected by the user, prompting connection of the selected at least one other device with a first communication section and, after the selected at least one other device reconnects with the first communication section, executing the duplication processing to the selected at least one other device.

9. A non-transitory storage medium storing a program that is executable by a computer to execute information processing, the information processing comprising: displaying information of at least one other device connected to an information processing device, the information processing device being usable as a key of a vehicle; receiving a user selection selecting, from the displayed at least one other device, at least one other device to be operated as a key of the vehicle; executing duplication processing to each of the selected at least one other device to allow the selected at least one other device to be used as the key of the vehicle; displaying the at least one other device connected to the information processing device with any one or more of communication sections together with a type of a communication section connected with the at least one other device; and in a case in which at least one other device that is connected with a second communication section is selected by the user, prompting connection of the selected at least one other device with a first communication section and, after the selected at least one other device reconnects with the first communication section, executing the duplication processing to the selected at least one other device.
